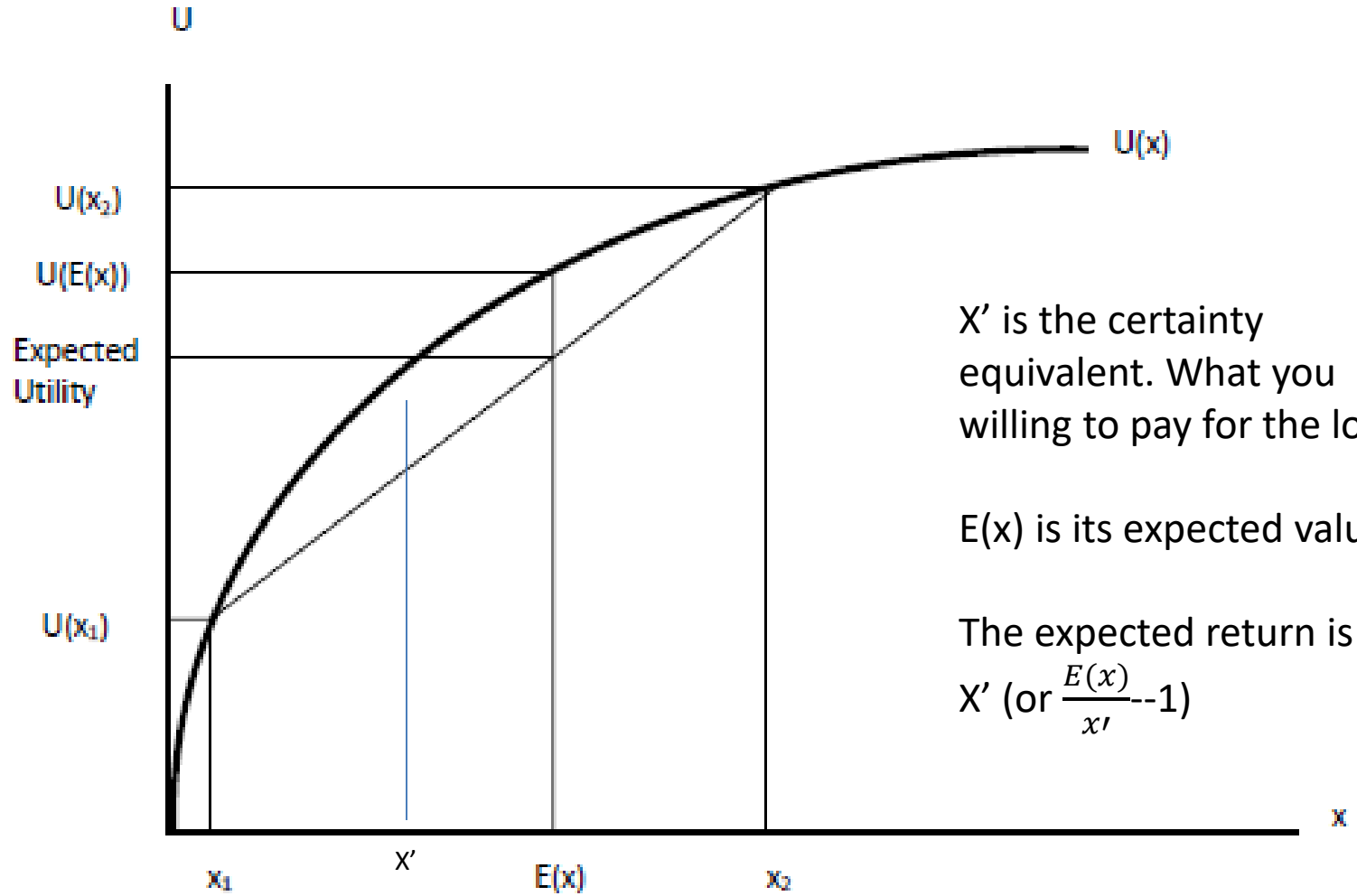
BEM 103

Class 08 : Mean vs Variance

- Mean variance preferences
- Basic mean variance trade off in utility;
- Why do we like to use this in finance
- Single assets vs portfolios

From Last time

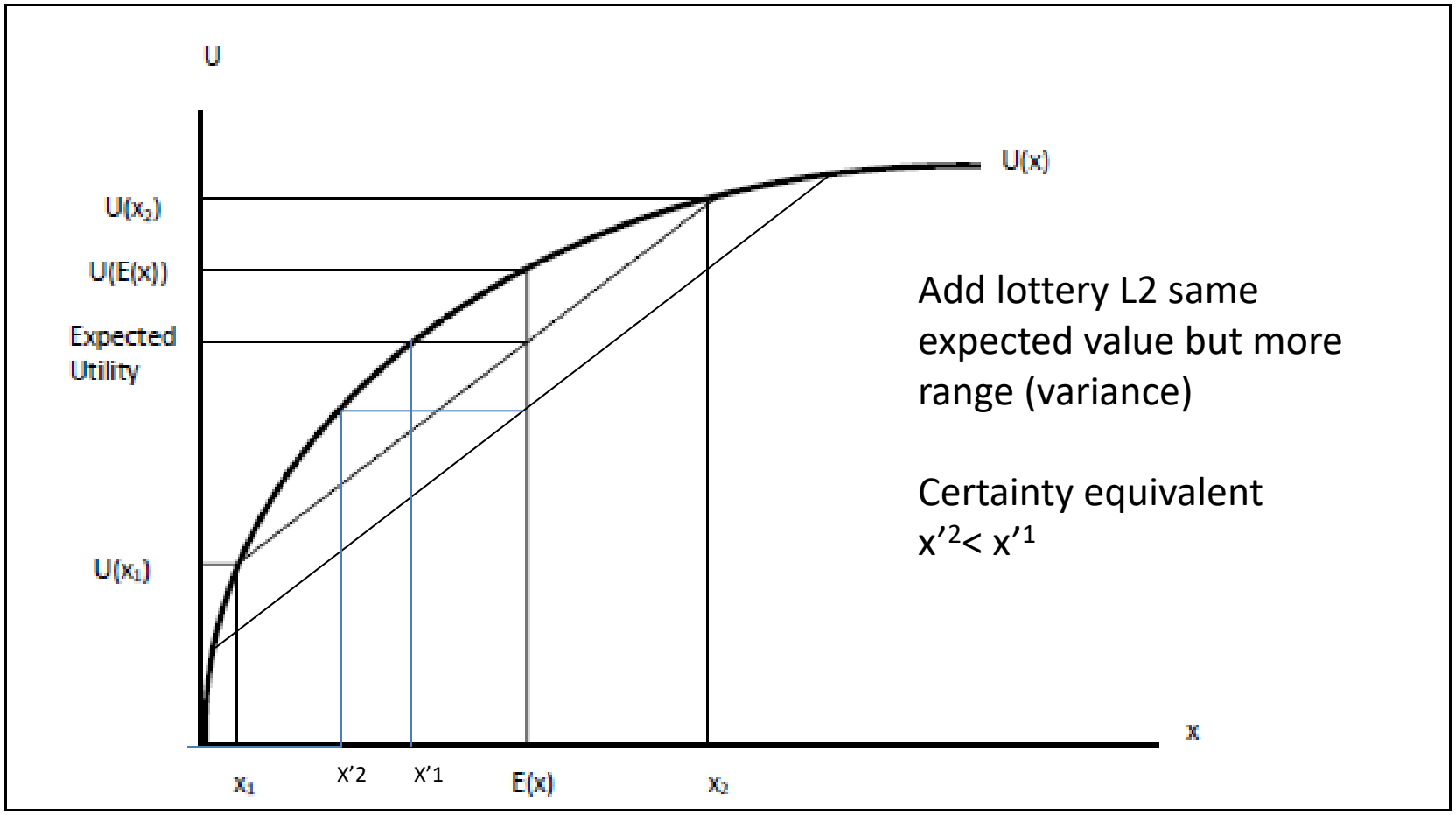


X' is the certainty equivalent. What you are willing to pay for the lottery

E(x) is its expected value

The expected return is E(x)-X' (or $\frac{E(x)}{x'}$--1)

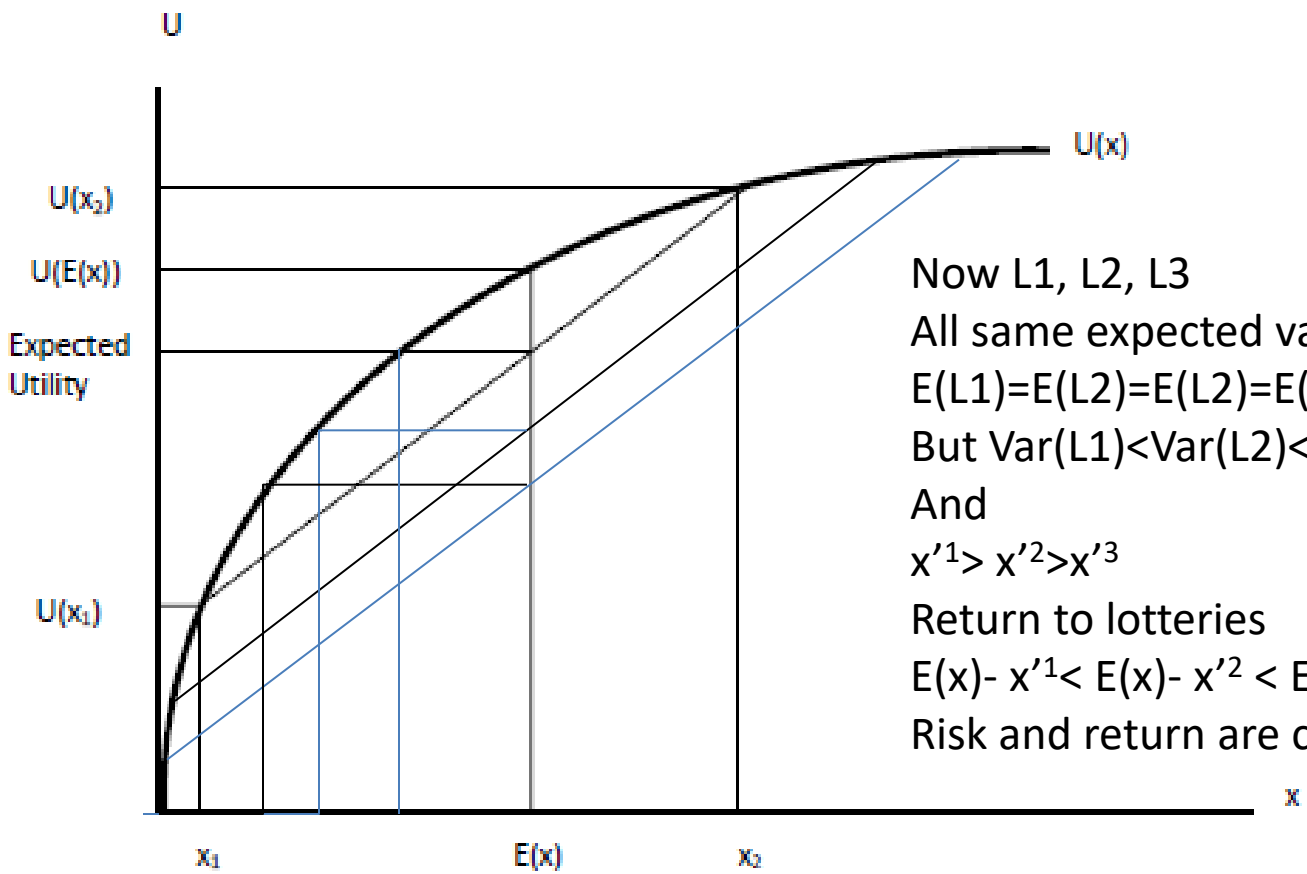
From Last time



Add lottery L2 same expected value but more range (variance)

Certainty equivalent $x'^2 < x'^1$

From Last time



Now L1, L2, L3

All same expected value

E(L1)=E(L2)=E(L2)=E(x)

But Var(L1)<Var(L2)<Var(L2)

And

$x'^1 > x'^2 > x'^3$

Return to lotteries

$E(x) - x'^1 < E(x) - x'^2 < E(x) - x'^3$

Risk and return are correlated!

Mean variance tradeoffs

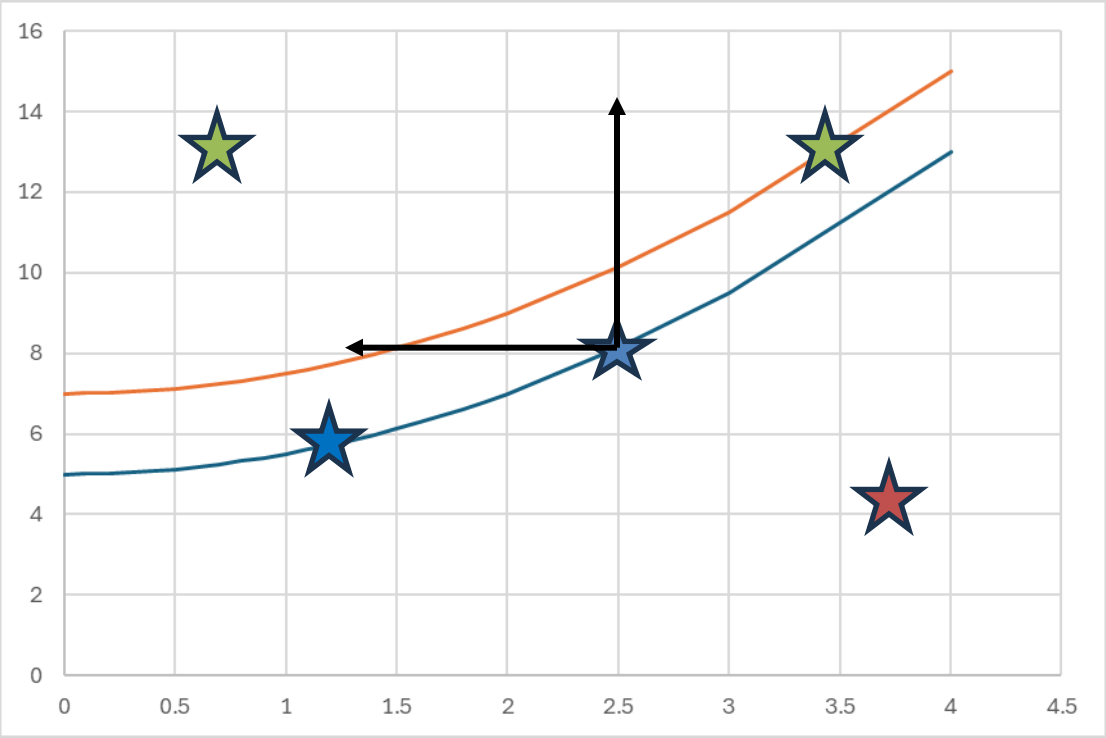
- Sample mean $E(x)$= $\sum_{i=1}^{n} \frac{x_i}{n}$ Sample Variance $var(x) = \sigma^2 = \sum_{i=1}^{n} \frac{(x_i - E(x))^2}{n-1}$
- Where i is the number of observation
- Mean $E(x)$= $\sum_{i=1}^{n} p_i x_i$ Variance $var(x) = \sigma^2 = \sum_{i=1}^{n} p_i (x_i - E(x))^2$

- Where p_i is the probability state of the world i occurs and x_i is the value of the security in state i
- Because utility is increasing in income we assume more mean return is better.
- Because the agent is risk averse we assume more variance is bad.

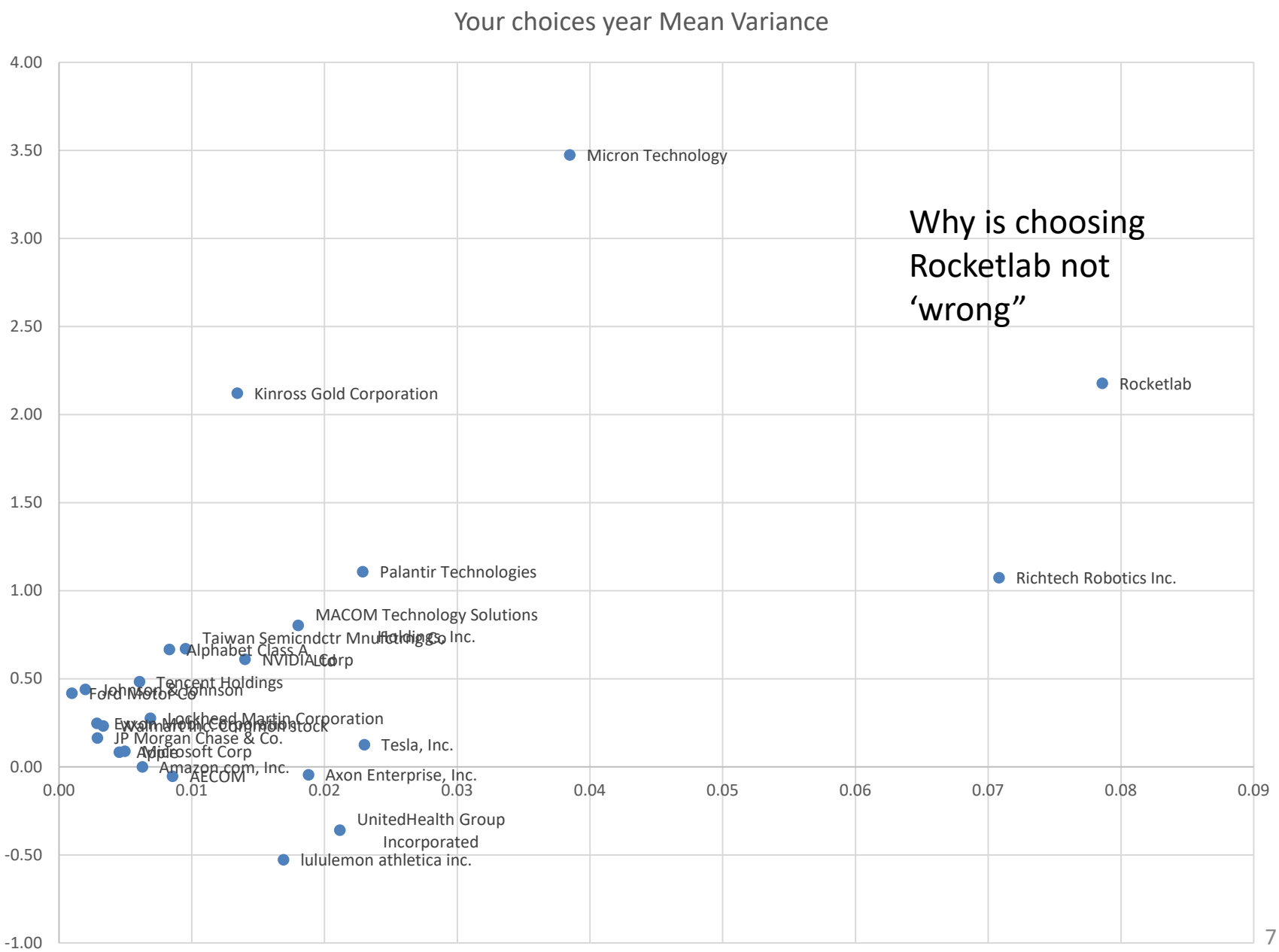
- That means the value function of an investor is like
- $V(x) = e(x) + bvar(x)$ or $V(x) = e(x) + bvar(x)^2$

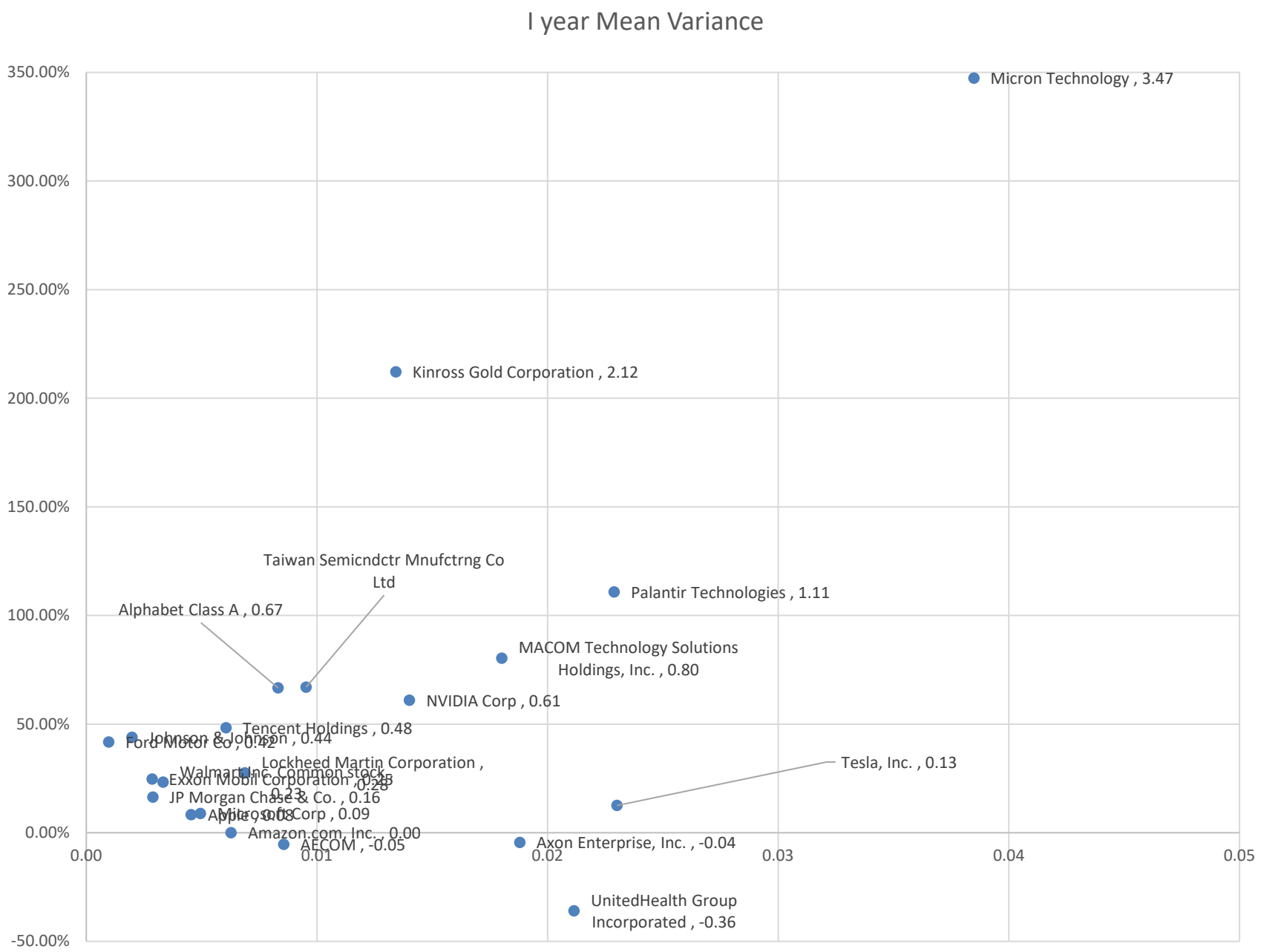
- We generally assume that b is negative (but if you are risk-loving then b will be positive, if risk neutral b=0).
- The more risk averse the larger b is.

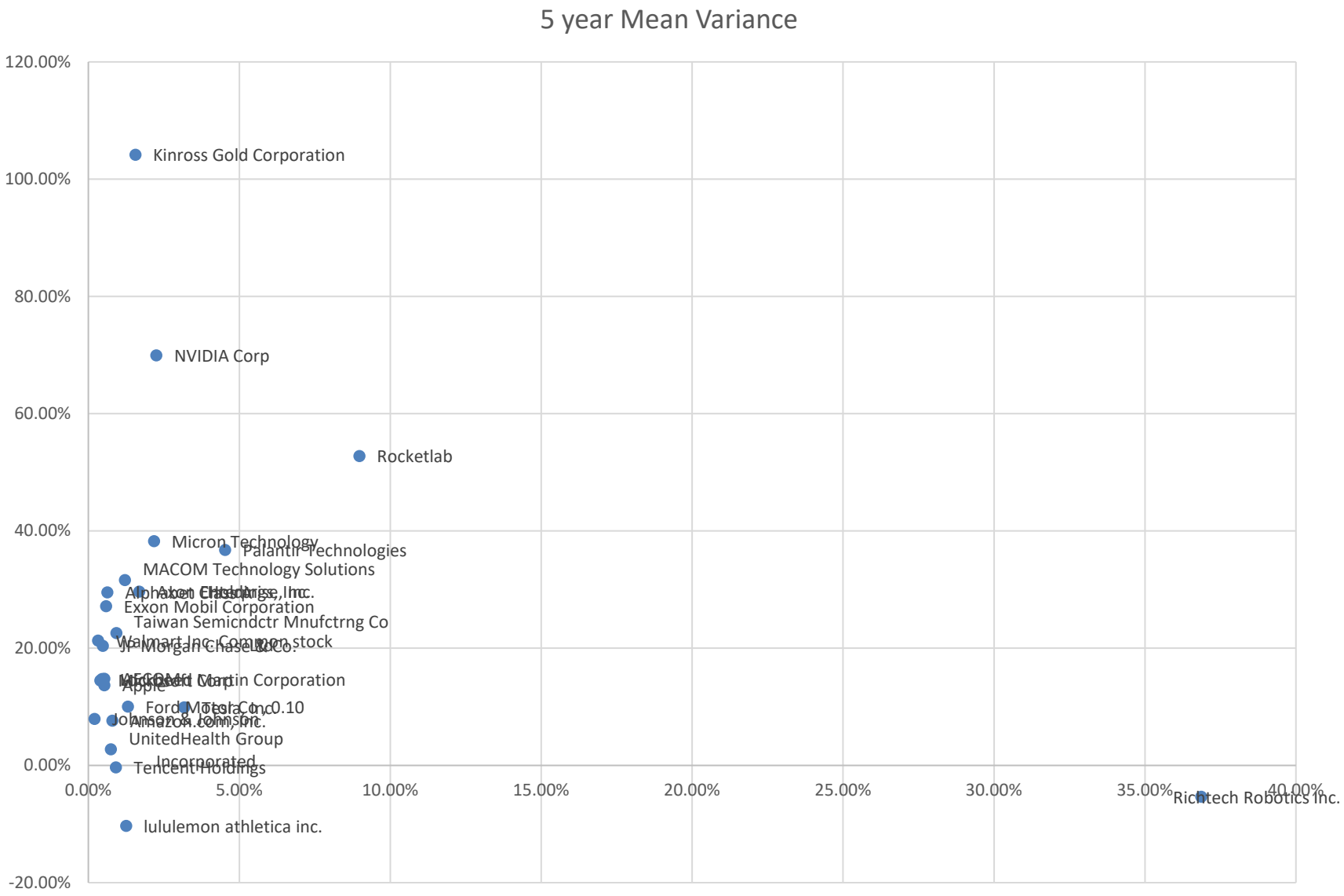
Mean variance tradeoff

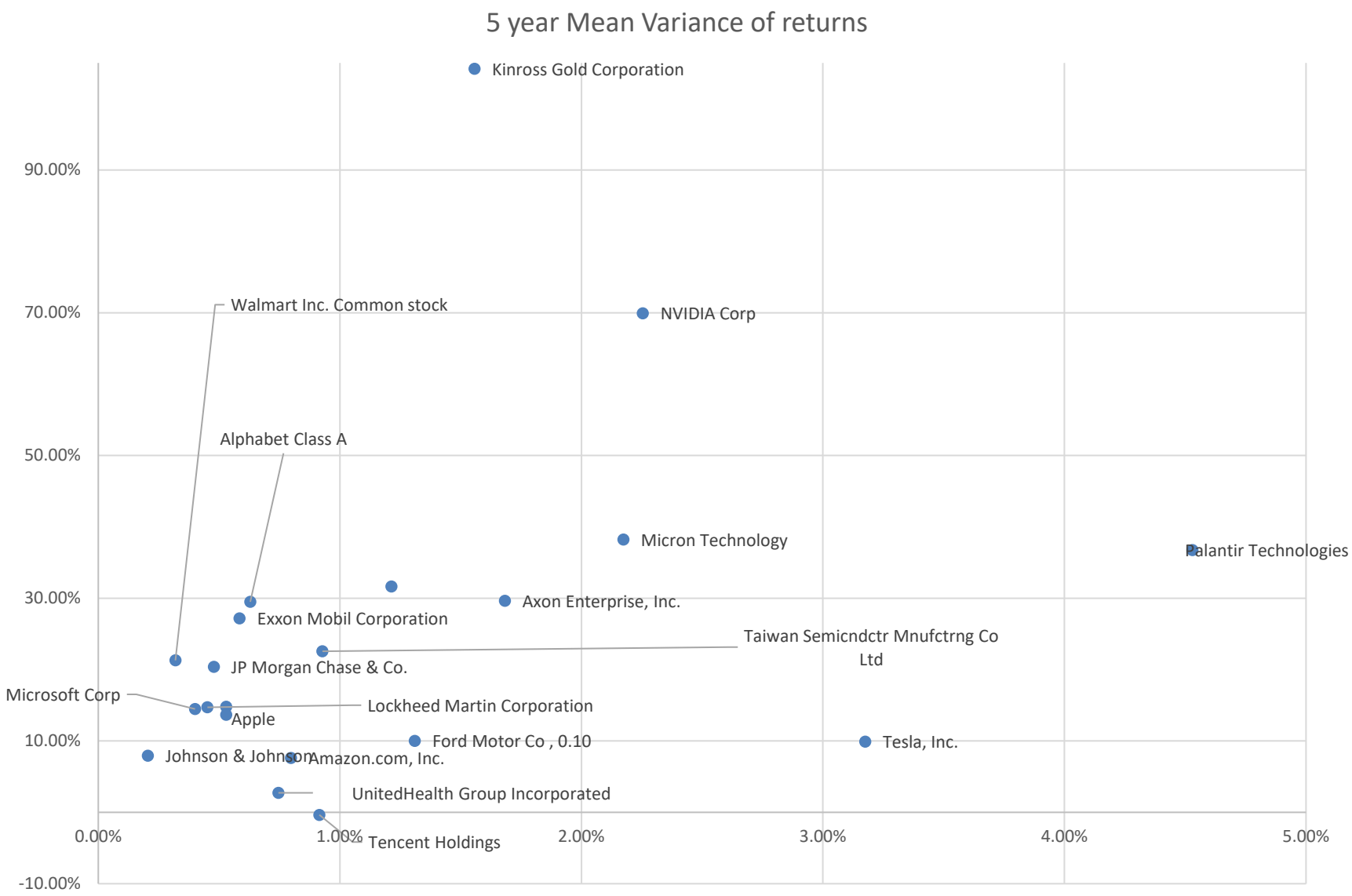


$$V(x) = e(x) + bvar(x)$$


Your choices year Mean Variance
4.00
3.50
3.00
2.50
2.00
1.50
1.00
0.50
0.00
-0.50
-1.00
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
Micron Technology
Why is choosing Rocketlab not 'wrong"
Rocketlab
Kinross Gold Corporation
Palantir Technologies
Richtech Robotics Inc.
MACOM Technology Solutions
Holdings, Inc.
Alphabet Class A
NVIDIA Corp
Tencent Holdings
Ford Motor Co
Lockheed Martin Corporation
JP Morgan Chase & Co.
Tesla, Inc.
Apple
Microsoft Corp
Amazon.com, Inc.
AECOM
Axon Enterprise, Inc.
UnitedHealth Group
Incorporated
lululemon athletica inc.


I year Mean Variance
350.00%
300.00%
250.00%
200.00%
150.00%
100.00%
50.00%
0.00%
-50.00%
0.00
0.01
0.02
0.03
0.04
0.05
Micron Technology , 3.47
Kinross Gold Corporation , 2.12
Taiwan Semicndctr Mnufctrng Co Ltd
Alphabet Class A , 0.67
Palantir Technologies , 1.11
MACOM Technology Solutions Holdings, Inc. , 0.80
NVIDIA Corp , 0.61
Tencent Holdings , 0.48
Johnson & Johnson , 0.44
Ford Motor Co , 0.42
Lockheed Martin Corporation , 0.28
Walmart Inc. Common stock , 0.23
Exxon Mobil Corporation , 0.23
JP Morgan Chase & Co. , 0.16
Apple , 0.08
Microsoft Corp , 0.09
Amazon.com, Inc. , 0.00
AECOM , -0.05
Tesla, Inc. , 0.13
Axon Enterprise, Inc. , -0.04
UnitedHealth Group Incorporated , -0.36


5 year Mean Variance
120.00%
100.00%
80.00%
60.00%
40.00%
20.00%
0.00%
-20.00%
0.00%
5.00%
10.00%
15.00%
20.00%
25.00%
30.00%
35.00%
40.00%
Kinross Gold Corporation
NVIDIA Corp
Rocketlab
Micron Technology
Palantir Technologies
MACOM Technology Solutions
Exxon Mobil Corporation
Taiwan Semicndctr Mnufctrng Co
Walmart Inc. Common stock
Apple
Ford Motor Co.
UnitedHealth Group
Tencent Holdings
Incorporated
lululemon athletica inc.
Richtech Robotics Inc.


5 year Mean Variance of returns
Kinross Gold Corporation
90.00%
70.00%
50.00%
30.00%
10.00%
-10.00%
Walmart Inc. Common stock
NVIDIA Corp
Alphabet Class A
Micron Technology
Palantir Technologies
Axon Enterprise, Inc.
Exxon Mobil Corporation
Taiwan Semicndctr Mnufctrng Co Ltd
JP Morgan Chase & Co.
Microsoft Corp
Lockheed Martin Corporation
Apple
Ford Motor Co , 0.10
Tesla, Inc.
Johnson & Johnson
Amazon.com, Inc.
UnitedHealth Group Incorporated
Tencent Holdings
0.00%
1.00%
2.00%
3.00%
4.00%
5.00%

Quiz time

From lotteries to assets

- Consider two lotteries with the same expected return of 50.
- S_1 .5 get 0 and .5 get 100 EV=.5*100=50
- S_2 .5 get 30 and .5 get 70 EV=.5*30+.5*70=15+35=50
- If you are risk averse you prefer S_2
- Another reason is that variance matters for the range of values your security can take over time (when you want to sell). If you are risk averse you prefer less variance.

The binomial Model

- Assume you have an asset
  - Equally likely to grow by a% or fall by b% from t to t+1
  - $E(A_{t+1}) = A_t[0.5(1+a) - 0.5(1-b)] = A_t[(1+0.5(a-b)]$
  - Growth rate (drift) is simply $0.5(a-b)$
  - Variance $\sigma^2\ (A_{t+1}) =$

$[A_t 0.5((1+a)-(a-b))]^2 + [A_t 0.5((1-b)-(a-b))]^2$

$$\sigma^2 = [0.5(1+b)]^2 + A_t[0.5(1-a)]^2 =$$

$$\sigma^2 = 0.25A_t^2\ [(1+b)^2 + (1-a)^2]$$

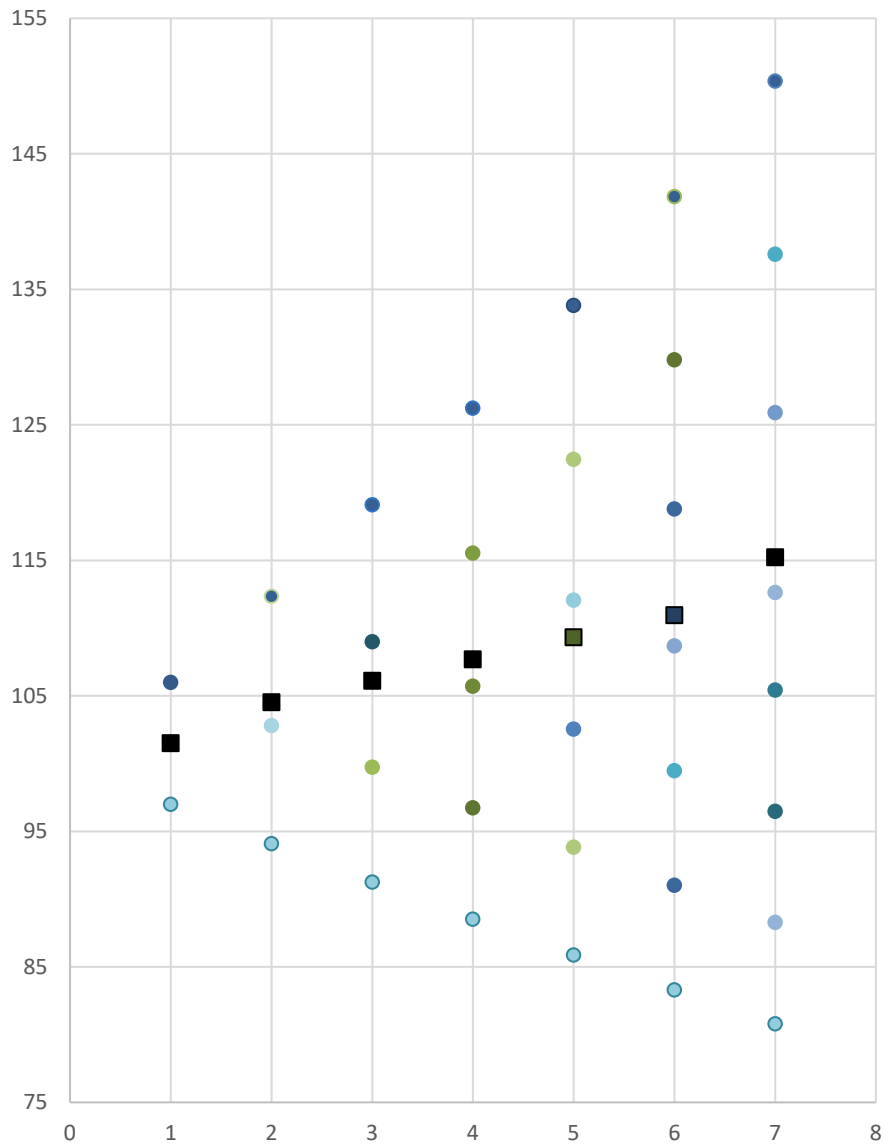
Normalized by A_t the variance is constant.

  - It's a simple practical model (and its continuous time limit is behind the core of options pricing)
  - Notice that holding b constant, increasing a increases drift.
  - Can increase b and a at the same time to increase variance leaving drift alone.

Assessing risk under Independence

- One more key assumptions of the model
  - Time independence (process is Markov)
  - Note all the assumption are wrong in practice but this is BEM103 we start with as simple
- Example
- You are a farmer. You own a tree that grows at rate r and thus yields (1.r)x fruit each year.
- The initial crop is 100 figs.
- Depending on whether it is a rainy or dry season the growth rate r is a=0.06 or b=-0.03.
- The probability of a rainy season is 0.5. So the expected growth of your fig harvest is 1.5% per year.
- What happens over time?

Growing the Tree



- So year 0 you get 100 figs
- Equally likely 106 or 97 figs.
- Variance is 40 (std dev 6.36 and coef of variation 6.3%)
- **Year 3**
- 25% probability 112 figs
- 50% probability 103 figs
- 25% probability 94 figs.
- The variance of the process after 7 years is large (173) deviation is 13 for a coefficient of variation of 11%).
- **After 7 years** with 22% you get less than 100 figs
- On the other hand, your expected return is 112 figs that year and with 33% you are getting more than 125 figs.
- That is the power of drift.

Drift vs independence

- Whatever the size of the tree year 6, equal probability of rain vs drought, so the past is irrelevant to the expected return henceforth.
- You do need to know the size of the tree and the likelihood of events (rain or drought)
- Suppose you had the worse outcome, (85 figs after 6 years).
  - Next year you are equally likely to go up to 90.1 (rain and growth at 6%) or down to 82.5 (dry and decline at 3%) mean expected growth 1.5% of an expected yield of 86.275.
  - That gives a variance of 29.26 (st dev 5.49 and coef of variation of 6.26)
  - But normalize for differences in initial conditions and the world is the same in each period.
- Reverse the logic.
- If the world is IID AND stable: then the mean return and the mean adjusted variance are forecastable from the data.
- Forecast the likely future states of the world AND price the security
- Certainty equivalent is mean-b*variance =86.275-b(29.26).

For you to check

- Depending on whether it is a rainy or dry season the growth rate r is 0.06 or -0.03, and expected growth rate is 0.015
- Alternative good better 0.07 bad worse-0.04. expected growth rate is 0.015
  - You can check (1) the range of tree size is larger in the second case and so is the variance.
  - Risk averse person prefers the tree we started with to the one on this slide (and for a given b) this tree will always be worth less given its current size.

Financial advisors in the single security world

- Measure mean and variance for each security
- Figure out each individual's risk tolerance (the b parameter
- Find each client the security that is on the highest indifference curve possible (that is the one they are willing to pay the most for).

We do not choose single securities

- Because they covary
- If I am risk averse I might want to add a ‘low’ return security that is badly correlated with my current portfolio because it will reduce variance.
- Start with one security then find another one that has good return and poor correlation => portfolio performance pb
- OR
- find two securities each with higher variance and higher return than the first one but taken together have higher return AND lower variance => arbitrage pb.

Portfolios

- A portfolio is simply a collection.
  - E.g. Past artistic achievement
  - Responsibilities (minister without portfolio!)
- A finance portfolio is thus a collection of assets.
  - These could be long positions—you own these assets and will enjoy the cash flow
  - They could be short—claims you promise to pay in the future
- The portfolio approach to finance:

Optimized collections of assets have better properties (lower variance, conditional on mean) than single assets because their variations offset through the covariance.

Next time

- <u>Class 09: The Portfolio approach to risk</u>
- **Portfolios have better mean return profiles than individual stocks;**
- **Efficient frontier and the Sharpe value;**
- **Basic portfolio separation;**
- **Why not insurance contracts?**